

TAREA T2: SISTEMAS NUMÉRICOS

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I) Realizar las conversiones siguientes de base 10 a base 2:

1.- $3576_{10} \rightarrow X_2$

3576	2
1788	0
0894	0
447	0
223	1
111	1
055	1
27	1
13	1
6	1
3	0
1	1
0	1

$3576_{10} \rightarrow 11011111000_2$

2.- $28,043_{10} \rightarrow X_2$

28	2
14	0
7	0
3	1
1	1
0	1

$0,043 \cdot 2 = 0,086$
$0,086 \cdot 2 = 0,172$
$0,172 \cdot 2 = 0,344$
$0,344 \cdot 2 = 0,688$
$0,688 \cdot 2 = 1,376$
$0,376 \cdot 2 = 0,752$
$0,752 \cdot 2 = 1,504$

$28,043_{10} \rightarrow 11100,0000101_2$

3.- $1357,2468_{10} \rightarrow X_2$

1357	2	$0,2468 \cdot 2 = 0,4936$
0678	1	$0,4936 \cdot 2 = 0,9872$
339	0	$0,9872 \cdot 2 = 1,9744$
169	1	$0,9744 \cdot 2 = 1,9488$
084	1	$0,9488 \cdot 2 = 1,8976$
42	0	$0,8976 \cdot 2 = 1,7952$
21	0	$0,7952 \cdot 2 = 1,5904$
10	1	$0,5904 \cdot 2 = 1,1808$
5	0	$0,1808 \cdot 2 = 0,3616$
2	1	$0,3616 \cdot 2 = 0,7232$
1	0	
0	1	

$1357,2468_{10} \rightarrow 10101001101,0011111100_2$

4.- $0,2589_{10} \rightarrow X_2$

$0,2589 \cdot 2 = 0,5178$
$0,5178 \cdot 2 = 1,0356$
$0,0356 \cdot 2 = 0,0712$
$0,0712 \cdot 2 = 0,1424$
$0,1424 \cdot 2 = 0,2848$
$0,2848 \cdot 2 = 0,5696$
$0,5696 \cdot 2 = 1,1392$
$0,1392 \cdot 2 = 0,2784$
$0,2784 \cdot 2 = 0,5568$
$0,5568 \cdot 2 = 1,1136$
$0,1136 \cdot 2 = 0,2272$

$0,2589_{10} \rightarrow 0,01000010010_2$

5.- $98,255_{10} \rightarrow X_2$

98	2	$0,255 \cdot 2 = 0,51$
49	0	$0,51 \cdot 2 = 1,02$
24	1	$0,02 \cdot 2 = 0,04$
12	0	$0,04 \cdot 2 = 0,08$
6	0	$0,08 \cdot 2 = 0,16$
3	0	$0,16 \cdot 2 = 0,32$
1	1	$0,32 \cdot 2 = 0,64$
0	1	$0,64 \cdot 2 = 1,28$
		$0,28 \cdot 2 = 0,56$

$98,255_{10} \rightarrow 1100010,010000010_2$

II) Realizar las conversiones siguientes de base 10 a otras bases:

6.- $123,571_{10} \rightarrow X_3$

123	3	$0,571 \cdot 3 = 1,713$
41	0	$0,713 \cdot 3 = 2,139$
13	2	$0,139 \cdot 3 = 0,417$
4	1	$0,417 \cdot 3 = 1,251$
1	1	$0,251 \cdot 3 = 0,753$
0	1	$0,753 \cdot 3 = 2,259$

$123,571_{10} \rightarrow 11120,120102_3$

7.- $628,68_{10} \rightarrow X_6$

628	6
104	4
017	2
5	5
0	2

0,68	*	6	=	4,08
0,08	*	6	=	0,48
0,48	*	6	=	2,88
0,88	*	6	=	5,28
0,28	*	6	=	1,68
0,68	*	6	=	4,08

$628,68_{10} \rightarrow 2524,402514_6$

8.- $2357,1468_{10} \rightarrow X_{16}$

2357	16
75	147
B7	(3)(9)
(5)	

0,1468	*	16	=	2,3488
0,3488	*	16	=	5,5808
0,5808	*	16	=	9,2928
0,2928	*	16	=	4,6848
0,6848	*	16	=	A,9568
0,9568	*	16	=	F,3088

$2357,1468_{10} \rightarrow 935,2594AF_{16}$

9.- $0,5871_{10} \rightarrow X_{11}$

0,5871	*	B(11)	=	6,4581
0,4581	*	B	=	5,0391
0,0391	*	B	=	0,4301
0,4301	*	B	=	4,7311
0,7311	*	B	=	8,0421
0,0421	*	B	=	0,4631
0,4631	*	B	=	5,0941

$0,5871_{10} \rightarrow 0,6504805_{11}$

10.- $258,97_{10} \rightarrow X_{20}$

$$\begin{array}{r} 258 \overline{) 20} \\ 58 \quad 12 \overline{) 20} \\ (1) \quad (12) \quad 0 \\ \quad \quad \quad c \end{array}$$

$$\begin{array}{l} 0,97 \star K(20) = 3,4 \\ 0,4 \star K = 8,0 \end{array}$$

$258,97_{10} \rightarrow CI, 38_{20}$

11.- $222,333_{10} \rightarrow X_4$

$$\begin{array}{r} 222 \overline{) 4} \\ 22 \quad 55 \overline{) 4} \\ (2) \quad 15 \quad 13 \overline{) 4} \\ \quad \quad (3) \quad (1) \quad (3) \end{array}$$

$$\begin{array}{l} 0,333 \star 4 = 1,332 \\ 0,332 \star 4 = 1,328 \\ 0,328 \star 4 = 1,312 \\ 0,312 \star 4 = 1,248 \\ 0,248 \star 4 = 0,992 \\ 0,992 \star 4 = 3,968 \end{array}$$

$222,333_{10} \rightarrow 3132,111103_4$

12.- $475,828_{10} \rightarrow X_{16}$

$$\begin{array}{r} 475 \overline{) 16} \\ F5 \quad 29 \overline{) 16} \\ (B) \quad (D) \quad 1 \end{array}$$

$$\begin{array}{l} 0,828 \star G(16) = 0,248 \\ 0,248 \star 16 = 3,968 \\ 0,968 \star 16 = F,488 \\ 0,488 \star 16 = 7,808 \\ 0,808 \star 16 = C,928 \end{array}$$

$475,828_{10} \rightarrow 10B,03F7C_{16}$

13.- $69,8459_{10} \rightarrow X_5$

69	5
13	4
2	3
0	2

$$0,8459 \cdot 5 = 4,2295$$

$$0,2295 \cdot 5 = 1,1475$$

$$0,1475 \cdot 5 = 0,7375$$

$$0,7375 \cdot 5 = 3,6875$$

$$0,6875 \cdot 5 = 3,4375$$

$$0,4375 \cdot 5 = 2,1875$$

$69,8459_{10} \rightarrow 234,410332_5$

III) Realizar las conversiones siguientes de base 2 a base 10

14.- $11101001111_2 \rightarrow X_{10}$

$$1 \cdot 2^{10} + 1 \cdot 2^9 + 1 \cdot 2^8 + \cancel{0 \cdot 2^7} + 1 \cdot 2^6 + \cancel{0 \cdot 2^5} + \cancel{0 \cdot 2^4} + 1 \cdot 2^3 + 1 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0$$

$$1024 + 512 + 256 + 64 + 8 + 4 + 2 + 1$$

$11101001111_2 \rightarrow 1871_{10}$

15.- $101100,11001_2 \rightarrow X_{10}$

$$1 \cdot 2^5 + \cancel{0 \cdot 2^4} + 1 \cdot 2^3 + 1 \cdot 2^2 + \cancel{0 \cdot 2^1} + \cancel{0 \cdot 2^0} + 1 \cdot 2^{-1} + 1 \cdot 2^{-2} + \cancel{0 \cdot 2^{-3}} + \cancel{0 \cdot 2^{-4}} + 1 \cdot 2^{-5}$$

$$32 + 8 + 4 + \frac{1}{2} + \frac{1}{4} + \frac{1}{32}$$

$$44 + 0,5 + 0,25 + 0,03125$$

$$44,78125$$

$101100,11001_2 \rightarrow 44,78125_{10}$

16.- $1001010,0110111_2 \rightarrow X_{10}$

$$1 \cdot 2^6 + 0 \cdot 2^5 + 0 \cdot 2^4 + 1 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 0 \cdot 2^0 + 0 \cdot 2^{-1} + 1 \cdot 2^{-2} + 1 \cdot 2^{-3} + 0 \cdot 2^{-4} + 1 \cdot 2^{-5} + 1 \cdot 2^{-6} + 1 \cdot 2^{-7}$$

$$64 + 8 + 2 + \frac{1}{4} + \frac{1}{8} + \frac{1}{32} + \frac{1}{64} + \frac{1}{128}$$

$$74 + 0,25 + 0,125 + 0,03125 + 0,015625 + 0,0078125$$

$$74,4296875$$

$1001010,0110111_2 \rightarrow 74,4296875_{10}$

17.- $1100010101,111_2 \rightarrow X_{10}$

$$2^9 + 2^8 + 2^4 + 2^2 + 2^0 + 2^{-1} + 2^{-2} + 2^{-3}$$

$$512 + 256 + 16 + 4 + 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8}$$

$$789 + 0,5 + 0,25 + 0,125$$

$1100010101,111_2 \rightarrow 789,875_{10}$

IV) Realizar las conversiones de diferentes bases a base 10:

18.- $4201,134_5 \rightarrow X_{10}$

$$4 \cdot 5^3 + 2 \cdot 5^2 + 0 \cdot 5^1 + 1 \cdot 5^0 + 1 \cdot 5^{-1} + 3 \cdot 5^{-2} + 4 \cdot 5^{-3}$$

$$500 + 50 + 1 + \frac{1}{5} + \frac{3}{25} + \frac{4}{125}$$

$$551 + 0,2 + 0,12 + 0,032 = 551,352_{10}$$

$4201,134_5 \rightarrow 551,352_{10}$

19.- $E3B12, C3D_{16} \rightarrow X_{10}$

$$E \cdot 16^1 + 3 \cdot 16^3 + B \cdot 16^2 + 1 \cdot 16^1 + 2 \cdot 16^0 + C \cdot 16^{-1} + 3 \cdot 16^{-2} + D \cdot 16^{-3}$$

$$917504 + 12288 + 2816 + 16 + 2 + \frac{12}{16} + \frac{3}{256} + \frac{13}{4096}$$

$$932626 + 0,75 + 0,01171875 + 0,003173828$$

$$E3B12, C3D_{16} \rightarrow 932626,765_{10}$$

20.- $2133,201_4 \rightarrow 10$

$$2 \cdot 4^3 + 1 \cdot 4^2 + 3 \cdot 4^1 + 3 \cdot 4^0 + 2 \cdot 4^{-1} + 0 \cdot 4^{-2} + 1 \cdot 4^{-3}$$

$$128 + 16 + 12 + 3 + \frac{2}{4} + \frac{1}{64}$$

$$159 + 0,5 + 0,015625$$

$$2133,201_4 \rightarrow 159,515625_{10}$$

21.- $4532,1234_6 \rightarrow X_{10}$

$$4 \cdot 6^3 + 5 \cdot 6^2 + 3 \cdot 6^1 + 2 \cdot 6^0 + 1 \cdot 6^{-1} + 2 \cdot 6^{-2} + 3 \cdot 6^{-3} + 4 \cdot 6^{-4}$$

$$864 + 180 + 18 + 2 + \frac{1}{6} + \frac{2}{36} + \frac{3}{216} + \frac{4}{1296}$$

$$1064 + 0,1666\bar{6} + 0,05\bar{5} + 0,013\bar{8} + 0,00308\bar{6}$$

$$4532,1234_6 \rightarrow 1064,23919628_{10}$$

V) Realizar las conversiones siguientes entre bases cualesquiera:

22.- $44,33_5 \rightarrow X_{16}$

$$4 \cdot 5^2 + 4 \cdot 5^1 + 3 \cdot 5^0 + 3 \cdot 5^{-1} + 3 \cdot 5^{-2}$$

$$20 + 4 + \frac{3}{5} + \frac{3}{25}$$

$$24 + 0,6 + 0,12$$

$$24,72$$

$$\begin{array}{r} 24 \overline{) 16} \rightarrow 18 \\ (8) \quad 1 \end{array}$$

$$0,72 \cdot 16 = 11,52$$

$$0,52 \cdot 16 = 8,32$$

$$0,32 \cdot 16 = 5,12$$

$$0,12 \cdot 16 = 1,92$$

$$0,92 \cdot 16 = 14,72$$

$44,33_5 \rightarrow 18,1851E_{16}$

23.- $8202,9F3_{16} \rightarrow X_7$

$$8 \cdot 16^3 + 2 \cdot 16^2 + 0 \cdot 16^1 + 2 \cdot 16^0 + 9 \cdot 16^{-1} + F \cdot 16^{-2} + 3 \cdot 16^{-3}$$

$$32768 + 512 + 2 + \frac{9}{16} + \frac{15}{256} + \frac{3}{4096}$$

$$33282 + 0,5625 + 0,058593 + 0,00073242$$

$$33282,62182(10)$$

$$\begin{array}{r} 33282 \overline{) 7} \end{array}$$

$$52$$

$$4754$$

$$38$$

$$55$$

$$679$$

$$7$$

$$32$$

$$64$$

$$49$$

$$97$$

$$7$$

$$(4)$$

$$(1)$$

$$(0)$$

$$27$$

$$13$$

$$7$$

$$(6)$$

$$(6)$$

$$1$$

$$0,62182 \cdot 7 = 4,3527$$

$$0,3527 \cdot 7 = 2,4689$$

$$0,4689 \cdot 7 = 3,2823$$

$$0,2823 \cdot 7 = 1,9761$$

$$0,9761 \cdot 7 = 6,8327$$

$8202,9F3_{16} \rightarrow 166014,42316_7$

24. $2232,3011_4 \rightarrow X_3$

$$2 \cdot 4^3 + 2 \cdot 4^2 + 3 \cdot 4^1 + 2 \cdot 4^0 + 3 \cdot 4^{-1} + 0 \cdot 4^{-2} + 1 \cdot 4^{-3} + 1 \cdot 4^{-4}$$

$$128 + 32 + 12 + 2 + \frac{3}{4} + \frac{1}{64} + \frac{1}{256}$$

$$174 + 0,75 + 0,015625 + 0,00390625$$

$$174,76953125_{10}$$

$$\begin{array}{r} 174 \quad \underline{3} \\ 24 \quad 58 \quad \underline{3} \\ (0) \quad 28 \quad 19 \quad \underline{3} \\ \quad (1) \quad (1) \quad 6 \quad \underline{3} \\ \quad \quad (0) \quad 2 \end{array}$$

$$\begin{array}{l} 0,76953 \cdot 3 = 2,30859 \\ 0,30859 \cdot 3 = 0,92577 \\ 0,92577 \cdot 3 = 2,77731 \\ 0,77731 \cdot 3 = 2,33193 \\ 0,33193 \cdot 3 = 0,99579 \end{array}$$

$2232,3011_4 \rightarrow 20110,20220_3$

25. $7132,23_6 \rightarrow X_8$

$$7 \cdot 6^3 + 1 \cdot 6^2 + 3 \cdot 6^1 + 2 \cdot 6^0 + 2 \cdot 6^{-1} + 3 \cdot 6^{-2}$$

$$5103 + 36 + 18 + 2 + \frac{2}{6} + \frac{3}{36}$$

$$5213 + 0,2 + 0,037037 \rightarrow 5213,259259_{10}$$

$$\begin{array}{r} 5213 \quad \underline{8} \\ 41 \quad 651 \quad \underline{8} \\ 13 \quad 11 \quad 81 \quad \underline{8} \\ (5) \quad (3) \quad (01) \quad 40 \quad \underline{8} \\ \quad \quad \quad (2) \quad 1 \end{array}$$

$$\begin{array}{l} 0,259259 \cdot 8 = 2,074072 \\ 0,074072 \cdot 8 = 0,592576 \\ 0,592576 \cdot 8 = 4,740608 \\ 0,740608 \cdot 8 = 5,924864 \\ 0,924864 \cdot 8 = 7,398912 \end{array}$$

$7132,23_6 \rightarrow 12135,20457$

26.- $456,348 \rightarrow x_{12}$

$$4 \cdot 8^2 + 5 \cdot 8^1 + 6 \cdot 8^0 + 3 \cdot 8^{-1} + 4 \cdot 8^{-2}$$

$$256 + 40 + 6 + \frac{3}{8} + \frac{4}{64}$$

$$302 + 0,375 + 0,0625$$

$$302,4375_{10}$$

302	12	
62	25	12
(2)	(1)	2

0,4375	12	=	5,25
0,25	12	=	3

$456,348 \rightarrow 212,53_{12}$